RadixMaths

Practice Paper 1: Mark Scheme and Worked Solutions

Pure Mathematics • Level: Standard

How to use this mark scheme

Each question has a **mark scheme** in the style the exam boards use, then a **worked solution** that explains each step. Mark your own work with the scheme first, then read the worked solution for anything you missed.

M	Method mark: a correct method applied to the problem, even if there are small slips.
dM	Dependent method mark: needs the method mark before it.
A	Accuracy mark: a correct answer from correct working. Needs the method mark it follows.
B	Independent mark for a correct result or statement.
ft	Follow through: allows an earlier wrong answer to be used correctly.
cso	Correct solution only: every step must be correct, with no errors.
awrt	Answers which round to the value given.
oe	Or equivalent: any correct equivalent form is accepted.

For "show that" questions the answer is given, so every step must be shown for full marks. Accept any correct method that is not in the scheme, and give the marks it would earn.

Question 1

Part	Scheme	Marks	Notes
	$1 - \dfrac{9\theta^2}{2}$	B1	
		(1)	

Worked solution

The small angle approximation is $\cos x \approx 1 - \dfrac{x^2}{2}$. Replace x with 3θ:

$$\cos 3\theta \approx 1 - \frac{(3\theta)^2}{2} = 1 - \frac{9\theta^2}{2}$$

Watch out: square the whole of 3θ. Writing $1 - \frac{3\theta^2}{2}$ squares only the θ.

Question 2

Part	Scheme	Marks	Notes
	Write as powers: $6x^2 - 4x^{-3} + 3x^{1/2}$, and raise the power of at least one term by 1	M1	
	$2x^3$	A1	
	$+2x^{-2}$	A1	oe, such as $+\dfrac{2}{x^2}$
	$+2x^{3/2} + c$	A1	Needs $+c$
		(4)	

Worked solution

Rewrite every term as a power of x first: $\dfrac{4}{x^3} = 4x^{-3}$ and $3\sqrt{x} = 3x^{1/2}$. Then add 1 to each power and divide by the new power:

$$\int \left(6x^2 - 4x^{-3} + 3x^{1/2}\right) \mathrm{d}x = \frac{6x^3}{3} - \frac{4x^{-2}}{-2} + \frac{3x^{3/2}}{3/2} + c$$
$$= 2x^3 + 2x^{-2} + 2x^{3/2} + c$$

Watch out: dividing by a negative power: $-4 \div (-2) = +2$.

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Part	Scheme	Marks	Notes

Question 3

Part	Scheme	Marks	Notes
(a)	$f(3) = 2(27) - 3(9) - 11(3) + 6$	M1	Attempt $f(3)$
	$= 54 - 27 - 33 + 6 = 0$, so $(x - 3)$ is a factor	A1	Needs $= 0$ and a conclusion
(b)	Divide or compare coefficients: $(x - 3)(2x^2 + 3x - 2)$	M1	
	$(x - 3)(2x - 1)(x + 2)$	A1	
(c)	$e^t = 3$ or $e^t = \frac{1}{2}$, rejecting $e^t = -2$	M1	Links x to e^t
	$t = \ln 3,\ t = -\ln 2$	A1	Accept $\ln \frac{1}{2}$. No extra answers
		(6)	

Worked solution

(a) By the factor theorem, $(x - 3)$ is a factor exactly when $f(3) = 0$.

$$f(3) = 54 - 27 - 33 + 6 = 0 \quad \Rightarrow \quad (x - 3) \text{ is a factor.}$$

(b) Divide $f(x)$ by $(x - 3)$ to get $2x^2 + 3x - 2$, which factorises:

$$f(x) = (x - 3)(2x^2 + 3x - 2) = (x - 3)(2x - 1)(x + 2)$$

(c) The equation is $f(e^t) = 0$, since $e^{3t} = (e^t)^3$ and $e^{2t} = (e^t)^2$. So $e^t = 3$, $\frac{1}{2}$ or -2. But $e^t > 0$ for every t, so $e^t = -2$ has no solution.

$$t = \ln 3 \qquad \text{or} \qquad t = \ln \tfrac{1}{2} = -\ln 2$$

Watch out: say why you reject $e^t = -2$. Examiners look for it.

Question 4

Part	Scheme	Marks	Notes
(a)	$2^6 = 64$	B1	
	$\binom{6}{1}2^5(-x)$ or $\binom{6}{2}2^4(-x)^2$	M1	Correct structure for either term
	$64 - 192x + 240x^2$	A1	
(b)	$3 \times 240 + 2 \times (-192)$	M1	Two products for x^2
	336	A1	
		(5)	

Worked solution

(a) Using $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots$ with $a = 2$, $b = -x$:

$$(2 - x)^6 = 2^6 + 6(2^5)(-x) + 15(2^4)(-x)^2 + \cdots = 64 - 192x + 240x^2 + \ldots$$

(b) An x^2 term comes from $3 \times 240x^2$ and from $2x \times (-192x)$:

$$720 - 384 = 336$$

Watch out: keep the minus sign inside the bracket, $(-x)$, so the signs alternate.

Question 5

Part	Scheme	Marks	Notes
(a)	$\log_2 \dfrac{x+1}{x} = 3$	M1	Subtraction law
	$\dfrac{x+1}{x} = 2^3 = 8$	M1	Removes logs correctly
	$x = \frac{1}{7}$	A1	
(b)	$(2x-1)\ln 5 = x \ln 7$	M1	Logs of both sides, any base
	$x(2\ln 5 - \ln 7) = \ln 5 \Rightarrow x = \dfrac{\ln 5}{2\ln 5 - \ln 7}$	M1	Collects the x terms
	$x = 1.26$	A1	awrt 1.26
		(6)	

Worked solution

(a) Combine the logs, then undo the log:

$$\log_2 \frac{x+1}{x} = 3 \;\Rightarrow\; \frac{x+1}{x} = 8 \;\Rightarrow\; x+1 = 8x \;\Rightarrow\; x = \frac{1}{7}$$

Check: $x = \frac{1}{7}$ makes x and $x+1$ positive, so both logs exist.
(b) Take natural logs of both sides and use $\ln a^k = k \ln a$:

$$(2x-1)\ln 5 = x \ln 7$$
$$2x \ln 5 - x \ln 7 = \ln 5$$
$$x = \frac{\ln 5}{2\ln 5 - \ln 7} = 1.264\ldots = 1.26 \text{ (3 s.f.)}$$

Question 6

Part	Scheme	Marks	Notes
(a)(i)	$(x-3)^2 + (y+2)^2 = 12 + 9 + 4$	M1	Completes the square for x and y
	Centre $(3, -2)$	A1	
(a)(ii)	Radius 5	A1	
(b)	Gradient of radius $= \dfrac{2-(-2)}{6-3} = \dfrac{4}{3}$	M1	
	Tangent gradient $-\frac{3}{4}$: $y - 2 = -\frac{3}{4}(x-6)$	M1	Perpendicular gradient, line through P
	$3x + 4y - 26 = 0$	A1	Accept any integer multiple
		(6)	

Worked solution

(a) Complete the square in x and in y:

$$(x-3)^2 - 9 + (y+2)^2 - 4 - 12 = 0 \;\Rightarrow\; (x-3)^2 + (y+2)^2 = 25$$

So the centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$.
(b) A tangent is perpendicular to the radius at the point of contact. The radius from $(3, -2)$ to $P(6, 2)$ has gradient $\frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$.

$$y - 2 = -\tfrac{3}{4}(x-6) \;\Rightarrow\; 4y - 8 = -3x + 18 \;\Rightarrow\; 3x + 4y - 26 = 0$$

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Question 7

Part	Scheme	Marks	Notes
(a)	$2r + 1.2r = 32$	M1	Perimeter = two radii + arc $r\theta$
	$3.2r = 32 \Rightarrow r = 10$	A1*	Given answer: no errors
(b)	$\frac{1}{2} \times 10^2 \times 1.2 = 60$ cm^2	B1	
(c)	Triangle $OAB = \frac{1}{2} \times 10^2 \times \sin 1.2$ $(= 46.60\ldots)$	M1	Calculator in radians
	$60 - 50\sin 1.2$	M1	Sector minus triangle
	13.4 cm^2	A1	awrt 13.4
		(6)	

Worked solution

(a) The perimeter is two radii plus the arc, and arc length = $r\theta$:

$$r + r + 1.2r = 32 \;\Rightarrow\; 3.2r = 32 \;\Rightarrow\; r = 10$$

(b) Sector area = $\frac{1}{2}r^2\theta = \frac{1}{2}(100)(1.2) = 60$ cm^2.
(c) The segment is the sector minus triangle OAB. Triangle area = $\frac{1}{2}ab\sin C$:

$$S = 60 - \tfrac{1}{2}(10)(10)\sin 1.2 = 60 - 46.60\ldots = 13.4 \text{ cm}^2 \text{ (3 s.f.)}$$

Watch out: your calculator must be in radians for $\sin 1.2$. In degrees you get $\sin 1.2°$, which is wrong.

Question 8

Part	Scheme	Marks	Notes
(a)	$\frac{dy}{dx} = 3x^2 - 12x + 9$	M1	Power reduced by 1
	$3(x-1)(x-3) = 0 \Rightarrow x = 1, 3$	M1	Sets = 0 and solves
	$(1, 6)$ and $(3, 2)$	A1	
	$\frac{d^2y}{dx^2} = 6x - 12$, value -6 at $x = 1$, 6 at $x = 3$	M1	Or any valid test
	$(1, 6)$ maximum, $(3, 2)$ minimum	A1	
(b)	Uses the y values of the turning points	M1	
	$2 < k < 6$	A1	Strict inequalities
		(7)	

Worked solution

(a) Stationary points are where $\frac{dy}{dx} = 0$:

$$3x^2 - 12x + 9 = 3(x-1)(x-3) = 0 \;\Rightarrow\; x = 1 \text{ or } x = 3$$

Substitute into the curve: $x = 1$ gives $y = 6$ and $x = 3$ gives $y = 2$.
The second derivative is $6x - 12$. At $x = 1$ it is $-6 < 0$, so $(1, 6)$ is a **maximum**. At $x = 3$ it is $6 > 0$, so $(3, 2)$ is a **minimum**.
(b) The solutions of the equation are where the horizontal line $y = k$ crosses the curve. A positive cubic rises to the maximum, falls to the minimum, then rises again, so the line crosses three times when it lies

strictly between them:

$$2 < k < 6$$

At $k = 2$ or $k = 6$ the line touches a turning point, giving only two distinct solutions.

Question 9

Part	Scheme	Marks	Notes
(a)	$k(3x-1)^4$	M1	Chain rule
	$15(3x-1)^4$	A1	
(b)	$2x\ln x + x^2 \times \dfrac{1}{x}$	M1	Product rule, two terms
	$2x\ln x + x$	A1	oe
(c)	$\dfrac{x \times 2\cos 2x - \sin 2x \times 1}{x^2}$	M1	Quotient rule, correct order
	$\dfrac{2x\cos 2x - \sin 2x}{x^2}$	A1	oe
		(6)	

Worked solution

(a) Chain rule: bring down the power, reduce it by 1, then multiply by the derivative of the bracket, which is 3.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 5(3x-1)^4 \times 3 = 15(3x-1)^4$$

(b) Product rule with $u = x^2$ and $v = \ln x$, so $u' = 2x$ and $v' = \frac{1}{x}$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x\ln x + x^2 \cdot \frac{1}{x} = 2x\ln x + x$$

(c) Quotient rule with $u = \sin 2x$ and $v = x$, so $u' = 2\cos 2x$ and $v' = 1$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{vu' - uv'}{v^2} = \frac{2x\cos 2x - \sin 2x}{x^2}$$

Question 10

Part	Scheme	Marks	Notes
(a)	$\mathrm{f}(x) > 3$	B1	
(b)	$y - 3 = \dfrac{2}{x-1} \Rightarrow x = 1 + \dfrac{2}{y-3}$	M1	Makes x the subject
	$\mathrm{f}^{-1}(x) = 1 + \dfrac{2}{x-3}$	A1	oe, such as $\dfrac{x-1}{x-3}$
	Domain $x > 3$	B1ft	Follows their range in (a)
(c)	$\mathrm{f}(3) = 4$	M1	Applies f first
	$\mathrm{g}(4) = 17$	A1	
		(6)	

Worked solution

(a) For $x > 1$, $\frac{2}{x-1}$ takes every positive value but is never 0. So $f(x) > 3$.
(b) Let $y = 3 + \frac{2}{x-1}$ and make x the subject:

$$y - 3 = \frac{2}{x-1} \Rightarrow x - 1 = \frac{2}{y-3} \Rightarrow x = 1 + \frac{2}{y-3}$$

So $f^{-1}(x) = 1 + \dfrac{2}{x-3}$. The domain of f^{-1} is the range of f, so it is $x > 3$.
(c) $gf(3)$ means apply f first: $f(3) = 3 + \frac{2}{2} = 4$. Then $g(4) = 16 + 1 = 17$.

Question 11

Part	Scheme	Marks	Notes
(a)	$2(1 - \sin^2 x) + 3\sin x = 3$	M1	Uses $\cos^2 x = 1 - \sin^2 x$
	$2\sin^2 x - 3\sin x + 1 = 0$	A1	
	$(2\sin x - 1)(\sin x - 1) = 0 \Rightarrow \sin x = \frac{1}{2},\ 1$	M1	Solves the quadratic in $\sin x$
	$x = 30°, 90°, 150°$	A1	All three, no extras
(b)	$2x - 30° = 60°$	M1	
	$2x - 30° = 240°$	M1	Adds 180° for a second value
	$x = 45°, 135°$	A1	Both, no extras
		(7)	

Worked solution

(a) Replace $\cos^2 x$ so the equation is in $\sin x$ only:

$$2 - 2\sin^2 x + 3\sin x = 3 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0$$

$\sin x = \frac{1}{2}$ gives $x = 30°$ or $180° - 30° = 150°$. $\sin x = 1$ gives $x = 90°$.
(b) If $0° \leqslant x < 180°$, then $-30° \leqslant 2x - 30° < 330°$. Tan repeats every 180°, so in this range

$$2x - 30° = 60° \text{ or } 240° \Rightarrow x = 45° \text{ or } 135°$$

Watch out: change the interval first, so you don't miss the second solution.

Question 12

Part	Scheme	Marks	Notes
(a)	$2 + 19 \times 0.5 = 11.5$ km	B1	
(b)	$S_{20} = \frac{20}{2}(2 \times 2 + 19 \times 0.5)$	M1	
	135 km	A1	
(c)	$\frac{n}{2}(4 + 0.5(n-1)) > 500 \Rightarrow n^2 + 7n - 2000 > 0$	M1	Sum formula > 500 (or = 500)
	$n = \dfrac{-7 + \sqrt{8049}}{2} = 41.36\ldots$	M1	Solves the quadratic
	Day 42	A1	Not 41
		(6)	

Worked solution

The distances form an arithmetic sequence with $a = 2$ and $d = 0.5$.
(a) $u_n = a + (n-1)d$, so $u_{20} = 2 + 19(0.5) = 11.5$ km.
(b) $S_n = \frac{n}{2}(2a + (n-1)d)$, so $S_{20} = 10(4 + 9.5) = 135$ km.
(c) Solve $S_n > 500$:

$$\frac{n}{2}(4 + 0.5(n-1)) > 500 \implies n(3.5 + 0.5n) > 1000 \implies n^2 + 7n - 2000 > 0$$

The positive root of $n^2 + 7n - 2000 = 0$ is $n = 41.36\ldots$, so the inequality first holds when $n = 42$. Check: $S_{41} = 492$ km, which is not enough, and $S_{42} = 514.5$ km. The answer is **day 42**.

Question 13

Part	Scheme	Marks	Notes
(a)	Two consecutive odd numbers $2k+1$ and $2k+3$; $(2k+1)^2 + (2k+3)^2$	M1	Correct general form
	$= 8k^2 + 16k + 10$	A1	
	$= 8(k^2 + 2k + 1) + 2$	M1	Separates a multiple of 8
	$k^2 + 2k + 1$ is an integer, so the sum is 2 more than a multiple of 8	A1	Needs a conclusion
(b)	$n = 3$: $2^3 + 1 = 9 = 3 \times 3$, not prime	B1	Any valid counter-example
		(5)	

Worked solution

(a) Let the odd numbers be $2k+1$ and $2k+3$, where k is an integer. Then

$$(2k+1)^2 + (2k+3)^2 = 4k^2 + 4k + 1 + 4k^2 + 12k + 9 = 8k^2 + 16k + 10 = 8(k^2 + 2k + 1) + 2$$

$k^2 + 2k + 1$ is an integer, so the sum is 2 more than a multiple of 8.
(b) Try small values: $n = 1$ gives 3 and $n = 2$ gives 5, both prime, but $n = 3$ gives $9 = 3 \times 3$, which is not prime. One counter-example is enough to show the statement is false.

Watch out: writing the numbers as n and $n + 2$ doesn't prove anything, because it doesn't use the fact that they are odd.

Question 14

Part	Scheme	Marks	Notes
(a)	$u = x$, $\frac{dv}{dx} = e^{2x}$: $\frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x}\,dx$	M1	Parts in the right direction
	$\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}$	A1	
	$+c$, all correct	A1	
(b)	$4x - x^2 = x \Rightarrow x = 0, 3$	B1	
	$\int (4x - x^2 - x)\,dx = \frac{3}{2}x^2 - \frac{1}{3}x^3$	M1	Integrates curve minus line
	$\left[\frac{3}{2}x^2 - \frac{1}{3}x^3\right]_0^3$	M1	Uses their limits
	$\frac{9}{2}$	A1	
		(7)	

Worked solution

(a) Integration by parts: $\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$. Choose $u = x$, so it becomes 1 when differentiated, and $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$, so $v = \frac{1}{2}\mathrm{e}^{2x}$:

$$\int x\mathrm{e}^{2x}\,\mathrm{d}x = \tfrac{1}{2}x\mathrm{e}^{2x} - \int \tfrac{1}{2}\mathrm{e}^{2x}\,\mathrm{d}x = \tfrac{1}{2}x\mathrm{e}^{2x} - \tfrac{1}{4}\mathrm{e}^{2x} + c$$

(b) The curve and line meet where $4x - x^2 = x$, so $x(3-x) = 0$ and $x = 0$ or 3. Between these the curve is above the line, so

$$\text{Area} = \int_0^3 \left(3x - x^2\right)\mathrm{d}x = \left[\tfrac{3}{2}x^2 - \tfrac{1}{3}x^3\right]_0^3 = \tfrac{27}{2} - 9 = \tfrac{9}{2}$$

Question 15

Part	Scheme	Marks	Notes
(a)	$\cos t = \frac{x}{3}$, $\sin t = \frac{y}{2}$ in $\cos^2 t + \sin^2 t = 1$	M1	
	$\frac{x^2}{9} + \frac{y^2}{4} = 1$	A1	oe, such as $4x^2 + 9y^2 = 36$
(b)	$\frac{\mathrm{d}x}{\mathrm{d}t} = -3\sin t$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2\cos t$	M1	
	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2\cos t}{-3\sin t}$ at $t = \frac{\pi}{3}$	M1	Divides the right way round
	$-\frac{2\sqrt{3}}{9}$	A1	oe, such as $-\frac{2}{3\sqrt{3}}$
(c)	$P\left(\frac{3}{2}, \sqrt{3}\right)$; $0 - \sqrt{3} = -\frac{2\sqrt{3}}{9}\left(x - \frac{3}{2}\right)$	M1	Tangent through P with $y = 0$
	$x = 6$	A1	
		(7)	

Worked solution

(a) $\cos t = \frac{x}{3}$ and $\sin t = \frac{y}{2}$. Since $\cos^2 t + \sin^2 t = 1$,

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \dfrac{2\cos t}{-3\sin t}$. At $t = \frac{\pi}{3}$, $\cos t = \frac{1}{2}$ and $\sin t = \frac{\sqrt{3}}{2}$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{-\frac{3\sqrt{3}}{2}} = -\frac{2}{3\sqrt{3}} = -\frac{2\sqrt{3}}{9}$$

(c) At $t = \frac{\pi}{3}$, P is $\left(\frac{3}{2}, \sqrt{3}\right)$. On the tangent, put $y = 0$:

$$-\sqrt{3} = -\frac{2\sqrt{3}}{9}\left(x - \frac{3}{2}\right) \Rightarrow x - \frac{3}{2} = \frac{9}{2} \Rightarrow x = 6$$

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Question 16

Part	Scheme	Marks	Notes
(a)	$f(2) = -3, f(3) = 13$	M1	Both values attempted
	Change of sign and f is continuous, so $2 < \alpha < 3$	A1	Needs both reasons
(b)	$x_2 = \sqrt[3]{11}$	M1	
	$x_2 = 2.2240, x_3 = 2.2684$	A1	Both correct
(c)	$f'(x) = 3x^2 - 3$	M1	
	$x_2 = 2.3 - \dfrac{0.267}{12.87}$	M1	Correct use of Newton–Raphson
	2.2793	A1	
		(7)	

Worked solution

(a) $f(2) = 8 - 6 - 5 = -3 < 0$ and $f(3) = 27 - 9 - 5 = 13 > 0$. f is continuous (it's a polynomial) and changes sign, so there is a root between 2 and 3.
(b) $x_2 = \sqrt[3]{3(2) + 5} = \sqrt[3]{11} = 2.2240$, and $x_3 = \sqrt[3]{3(2.22398\ldots) + 5} = 2.2684$.
Use the unrounded x_2 to find x_3. Calculators make this easy with the ANS key.
(c) The Newton–Raphson formula is $x_2 = x_1 - \dfrac{f(x_1)}{f'(x_1)}$, with $f'(x) = 3x^2 - 3$:

$$f(2.3) = 0.267, \quad f'(2.3) = 12.87, \quad x_2 = 2.3 - \frac{0.267}{12.87} = 2.2793$$

For comparison, $\alpha = 2.27902\ldots$, so one step of Newton–Raphson is already very close.

Question 17

Part	Scheme	Marks	Notes
(a)	$\displaystyle\int \frac{1}{\theta - 20}\,d\theta = \int -k\,dt$	M1	Separates the variables
	$\ln(\theta - 20) = -kt + c$	A1	
	$t = 0, \theta = 90 \Rightarrow c = \ln 70$	M1	Finds the constant
	$\theta - 20 = 70e^{-kt} \Rightarrow \theta = 20 + 70e^{-kt}$	A1*	Given answer: no errors
(b)	$55 = 20 + 70e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2}$	M1	
	$k = \frac{1}{5}\ln 2$	A1	oe exact
(c)	$30 = 20 + 70e^{-kt} \Rightarrow t = \dfrac{\ln 7}{k}$	M1	Uses their k
	14.0 minutes	A1	awrt 14.0
		(8)	

Worked solution

(a) Separate the variables and integrate:

$$\int \frac{1}{\theta - 20}\,d\theta = \int -k\,dt \;\Rightarrow\; \ln(\theta - 20) = -kt + c$$

The tea is always hotter than the room, so $\theta - 20 > 0$ and no modulus sign is needed. At $t = 0$, $\theta = 90$,

so $c = \ln 70$. Then

$$\ln\frac{\theta - 20}{70} = -kt \;\Rightarrow\; \theta = 20 + 70\mathrm{e}^{-kt}$$

(b) $55 = 20 + 70\mathrm{e}^{-5k}$, so $\mathrm{e}^{-5k} = \frac{35}{70} = \frac{1}{2}$. Then $-5k = \ln\frac{1}{2} = -\ln 2$, so $k = \frac{\ln 2}{5}$.

(c) $30 = 20 + 70\mathrm{e}^{-kt}$, so $\mathrm{e}^{-kt} = \frac{1}{7}$ and $t = \dfrac{\ln 7}{k} = \dfrac{5\ln 7}{\ln 2} = 14.0$ minutes.

End of mark scheme. Every answer here was checked with computer algebra.

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